

MacSync-WRTDX5 Wi-Fi RTD (PT1000) Temperature Sensor Datasheet

Product Overview

MacSync-WRTDX5 is a high-accuracy **Wi-Fi-based RTD (PT1000) temperature sensor** designed for precise and reliable temperature monitoring in industrial and commercial environments where direct IP connectivity is available. Supporting **PT1000 probes with IEC 60751 linearization**, it delivers stable and repeatable measurements across a wide temperature range, making it suitable for both low-temperature and high-temperature process applications.

Built with an industrial-grade analog front end, low-noise signal conditioning, and minimal long-term drift, MacSync-WRTDX5 ensures dependable performance in manufacturing lines, process plants, HVAC systems, cold-chain facilities, boilers, furnaces, and thermal equipment installed within Wi-Fi-covered premises.

The device transmits temperature data securely over standard **Wi-Fi networks (2.4 GHz / 5 GHz)** to **on-premise servers, SCADA systems, local dashboards, or cloud platforms** using industry-standard protocols such as **MQTT, HTTP, and HTTPS**, without requiring gateways or long-range wireless infrastructure.

MacSync-WRTDX5 supports **configurable sampling intervals, threshold-based alerts**, and **local BLE-based provisioning** via the Macnman Maya mobile application. Available in **battery-operated and DC-powered variants**, it offers flexible deployment options for fixed installations as well as locations where wired power is limited.

From **industrial process control and predictive maintenance** to **HVAC monitoring, food & pharmaceutical compliance, and equipment safety**, MacSync-WRTDX5 delivers **accurate, low-latency Wi-Fi temperature monitoring** for modern Industrial IoT deployments.



Key Features

High-Accuracy PT1000 RTD Input

Compatible with PT1000 probes using IEC 60751 linearization for precise and repeatable industrial temperature measurement.

Wi-Fi Based Connectivity

Direct temperature data transmission over standard Wi-Fi networks (2.4 GHz / 5 GHz) without gateways.

Industrial-Grade Signal Conditioning

Low-noise analog front end with excellent stability, low drift, and high repeatability.

Event-Based & Interval Temperature Alerts

Configurable threshold, interval, and event-based alerts for process safety and monitoring.

Wide Temperature Measurement Range

Supports $-200\text{ }^{\circ}\text{C}$ to $+850\text{ }^{\circ}\text{C}$, suitable for cryogenic, process, and high-temperature industrial applications.

Battery or DC Powered Operation

Available in battery-operated and externally powered variants for flexible industrial deployments.

Rugged & Ingress-Protected Enclosure

IP65 standard enclosure with optional IP67 variant for harsh industrial environments.

Standard IP Protocol Support

Supports MQTT, HTTP, and HTTPS for seamless integration with SCADA systems, dashboards, and cloud platforms.

Secure Wi-Fi Communication

WPA2 / WPA3 Wi-Fi security with TLS-encrypted data transmission.

SS304 / SS316 Probe Support

Corrosion-resistant stainless-steel probe options for chemical, food, and process industries.

Suited Applications

Industrial Process Temperature Monitoring

High-accuracy PT1000 temperature sensing for reactors, tanks, ovens, kilns, boilers, and continuous process equipment within Wi-Fi-enabled facilities.

Boilers, Furnaces & Thermal Systems

Reliable high-temperature monitoring for combustion control, thermal safety, and efficiency optimization.

Cold Chain & Refrigeration Systems

Temperature compliance monitoring for pharmaceutical storage, food processing units, cold rooms, and refrigerated warehouses.

Food, Pharma & Clean-Room Environments

Ensures temperature compliance in regulated environments such as pharma manufacturing, laboratories, and clean rooms.

Local BLE Configuration

Easy provisioning, diagnostics, and configuration using the Macman Maya mobile application.

Predictive Maintenance Ready

Enables early detection of abnormal temperature trends in motors, pumps, pipelines, and thermal assets.

Predictive Maintenance for Machines & Motors

Detects abnormal temperature rise in motors, bearings, pumps, compressors, and rotating equipment to prevent failures and downtime.

Pipelines & Heat Exchangers

Temperature monitoring for pipelines, heat exchangers, and process lines installed within industrial plants and utility rooms.

HVAC & Building Automation

Integrates with HVAC and BMS systems for thermal control, energy optimization, and indoor comfort monitoring.

Smart Factories & Industrial IoT

Enables real-time temperature visibility across machines and processes using direct Wi-Fi connectivity and IP-based protocols.

Technical Specifications

General Specifications

Part Order Number	MacSync-WRTDX5
Product Brand	Macman Technologies Pvt Ltd

Temperature Measurement

Sensor Type	RTD (PT1000)
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Range	–200 °C to +850 °C
Accuracy	±0.2 °C (typical)
Resolution	0.1 °C
Repeatability	±0.05 °C
Probe Options	Internal & External Both Options
Sensor Linearization	IEC 60751 standard curve
Self-Heating Error	< 0.1 °C @ 1 mA excitation
Drift (1 Year)	< 0.05 °C
Insulation Resistance	> 100 MΩ @ 500 VDC
Probe Material	SS304 / SS316 (optional)

Wireless Communication

Wi-Fi Standards	IEEE 802.11 b/g/n/ac
Frequency Bands	2.4 GHz & 5 GHz (Dual-Band)
Operating Modes	Station Mode (Client) Access Point Mode (Configuration)
Network Protocols	MQTT, HTTP, HTTPS
Security	WPA2-PSK / WPA3 TLS / SSL Encrypted Communication
IP Configuration	DHCP / Static IP
Typical Indoor Range	Up to 100 m (environment dependent)

LED Status

LED Type	1 Internal Multi-Color LED
Solid Blue	Device is connected with Maya App
Flashing Blue	OTA (Over-the-Air) update in progress
Slow Blinking Green	Attempting to connect to the network
Single Red Blink	A confirmed parameter of MacSync is being modified

Device Configuration

Configuration	Via Macnman Maya Android App
Uplink Modes	Time Based / Threshold Based / Sampling Based
Stored Records*	30,000 Records
Remote Configuration	JSON Format Data - from server

Supported Cloud & Platforms

Deployment Models	On-Premise Servers Private Cloud Deployments Public Cloud Platforms
IoT Cloud Platforms	AWS IoT Core Microsoft Azure IoT Hub Google Cloud (via MQTT / HTTPS) Custom IoT Platforms
Integration Methods	MQTT Brokers REST / HTTPS APIs Webhooks
Enterprise Systems	SCADA Systems BMS / IBMS Platforms Data Lakes & Analytics Engines

Physical Characteristics

Power Supply	9-36 V DC
Operating Temperature	-40°C to +80°C
Humidity Range	0% to 100% RH (non-condensing)
Ingress Protection	IP65 standard (IP67 optional variant)
Housing Material	Glass filled Nylon
Color	Matte White
Weight	~110g
Mounting Options	Pole Mounting / Wall Mounting

Electrical Characteristics

Input Voltage Range	9-36 V DC (External Supply)
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Power Consumption	~100 µA in sleep mode ~150 mA during transmission
Input Impedance	>10 MΩ for voltage inputs

Communication Options

BLE Connectivity	For local configuration and diagnostics via Maya App
Cloud Protocols	MQTT / HTTPS
Serial Interface	RS485 Modbus RTU support (optional)

Certifications & Compliance

WPC	Compliant
RoHS	Conforms to RoHS directive

Security & Encryption

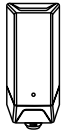
Data Encryption	AES-256 end-to-end encryption
Authentication	OTAA activation modes
Firmware Protection	Secure boot and signed OTA updates

MacSync-WRTDX5 Variants

VARIANT NAME	POWER SOURCE	BATTERY CAPACITY	SOLAR SUPPORT	EXTERNAL FLASH	IDEAL FOR
MacSync-WRTDX5-PO	12–24 V (Power)				Industrial or commercial sites with stable power supply
MacSync-WRTDX5-BO	Battery	19 Ah			Remote or portable deployments with low power

What's in the Box?

Make sure box contains following items before installations



MacSync



**Mounting
Clip**



**User manual &
Warranty Card**



**Junction
Box***



**Mounting
Screws**

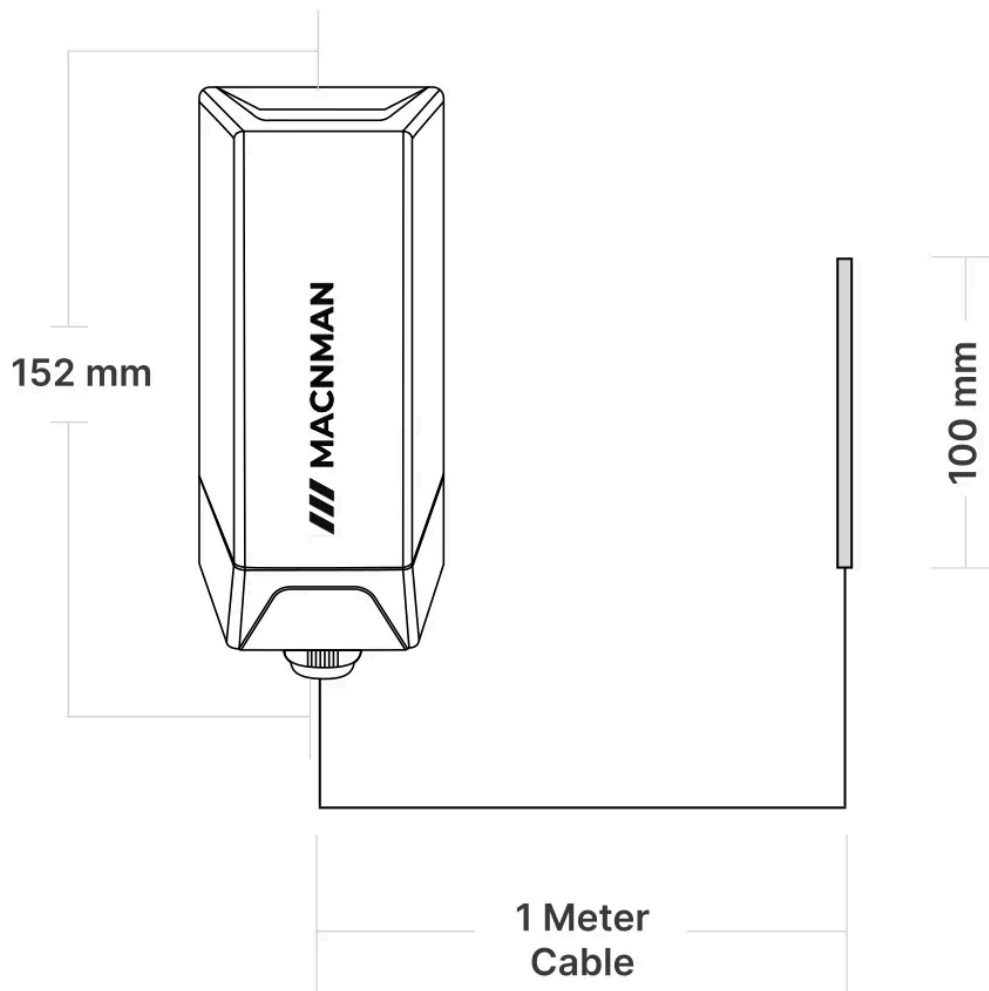


**Power
Adapters***

** Note**

Accessories must be purchased separately & may not be included in the box.

MacSync-WRTDX5 Dimensions



Frequently Asked Questions

What is the MacSync-WRTDX5 used for?

MacSync-WRTDX5 is a high-accuracy Wi-Fi RTD (PT1000) temperature sensor for precise, reliable temperature monitoring in industrial environments. Typical applications include Industrial Process Temperature Monitoring, Predictive Maintenance for Machines & Motors, Boilers, Furnaces & Thermal Systems, Pipelines & Heat Exchangers, Cold Chain & Refrigeration Systems and HVAC & Building Automation.

What does the MacSync-WRTDX5 measure, and how accurately?

Measurement specifications of the MacSync-WRTDX5:

- Temperature: range $-200\text{ }^{\circ}\text{C}$ to $+850\text{ }^{\circ}\text{C}$, accuracy $\pm 0.2\text{ }^{\circ}\text{C}$ (typical), resolution $0.1\text{ }^{\circ}\text{C}$.

How does the MacSync-WRTDX5 connect and send data over Wi-Fi?

The MacSync-WRTDX5 connects over IEEE 802.11 b/g/n/ac Wi-Fi on 2.4 GHz & 5 GHz (Dual-Band).

Security: WPA2-PSK / WPA3, TLS / SSL Encrypted Communication. IP configuration: DHCP / Static IP. It sends data using MQTT, HTTP, HTTPS. Typical indoor range is up to 100 m (environment dependent).

How is the MacSync-WRTDX5 powered?

The MacSync-WRTDX5 is powered from 9-36 V DC (External Supply).

What is the IP rating and operating temperature of the MacSync-WRTDX5?

The MacSync-WRTDX5 has an ingress protection rating of IP65 standard (IP67 optional variant) and an operating temperature range of -40°C to $+80^{\circ}\text{C}$. Operating humidity: 0% to 100% RH (non-condensing).

Mounting options: Pole Mounting / Wall Mounting.

How is the MacSync-WRTDX5 configured?

The MacSync-WRTDX5 is configured with the Macnman Maya Android app. Supported uplink modes: Time Based / Threshold Based / Sampling Based. It can also be configured remotely with JSON data sent from the server.

Which cloud platforms does the MacSync-WRTDX5 work with?

The MacSync-WRTDX5 can send data to platforms including AWS IoT Core, Microsoft Azure IoT Hub, Google Cloud (via MQTT / HTTPS), Custom IoT Platforms. Deployment models: On-Premise Servers, Private Cloud Deployments, Public Cloud Platforms. Integration methods: MQTT Brokers, REST / HTTPS APIs, Webhooks.

Related Products and Resources

- Related datasheets: [METH 301 compact Wi-Fi temperature and humidity sensor](#), [MacSync-WT1 Wi-Fi temperature and humidity sensor](#), [MacSync-WRTDX2 Wi-Fi RTD PT100 temperature sensor](#)
- Help: [Warranty terms and conditions](#) and [contact Macnman support](#)